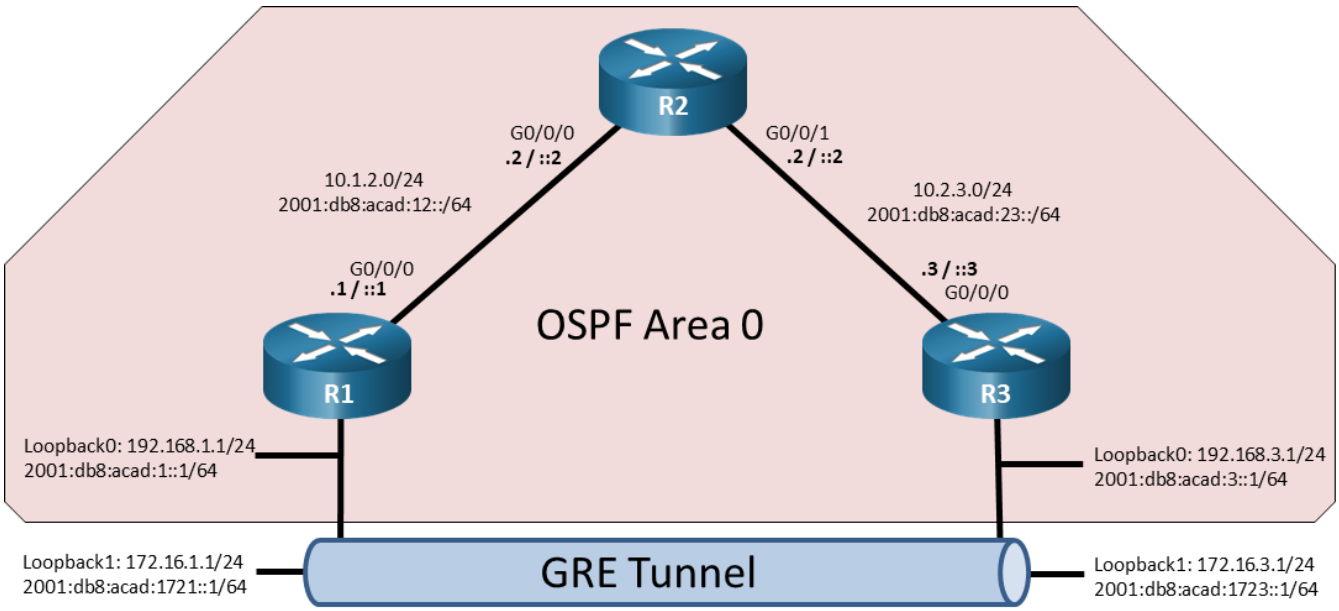
Lab - Implement a GRE Tunnel

Topology



Addressing Table

Device	Interface	IPv4 Address	IPv6 Address	IPv6 Link-Local
R1	G0/0/0	10.1.2.1/24	2001:db8:acad:12::1/64	fe80::1:1
	Loopback 0	192.168.1.1/24	2001:db8:acad:1::1/64	fe80::1:2
	Loopback 1	172.16.1.1/24	2001:db8:acad:1721::1/64	fe80::1:3
R2	G0/0/0	10.1.2.2/24	2001:db8:acad:12::2/64	fe80::2:1
	G0/0/1	10.2.3.2/24	2001:db8:acad:23::2/64	fe80::2:1
R3	G0/0/0	10.2.3.3/24	2001:db8:acad:23::3/64	fe80::3:1
	Loopback 0	192.168.3.1/24	2001:db8:acad:3::1/64	fe80::3:2
	Loopback 1	172.16.3.1/24	2001:db8:acad:1723::1/64	fe80::3:3

Objectives

Part 1: Build the Network and Configure Basic Device Settings

Part 2: Configure and Verify GRE Tunnels with Static Routing

Part 3: Configure and Verify GRE Tunnels by Using a Routing Protocol

Part 4: Examine the Recursive Routing Problem with GRE

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

Background / Scenario

Overlay networks are virtual networks that are built on top of existing physical or logical network infrastructures, which are known as underlay networks. By creating overlay networks, you can introduce additional flexibility and features to your network design without having to modify the underlying physical topology. This approach allows for easier implementation of new services, improved scalability, and better control over network traffic.

One key technology that enables the creation of overlay networks is Cisco's Generic Routing Encapsulation (GRE) protocol. GRE is a tunneling protocol that allows you to encapsulate a wide variety of network layer protocols inside virtual point-to-point links, which are then transmitted across the underlay network. This means you can use GRE tunnels to connect remote sites, transport non-IP traffic, or create secure communication channels, regardless of the underlying network's protocol.

GRE is highly flexible and supports both IPv4 and IPv6 as underlay networks, making it a versatile tool for many different networking scenarios. In this lab, you will learn how to deploy basic GRE tunnels over both IPv4 and IPv6 underlay networks.

Topology Note: In the topology, the **OSPF Area 0** network, encompassing R1, R2, R3, and their interconnected links, functions as the **underlay** network, providing the essential physical and logical IP connectivity. Despite its visual placement, the **GRE Tunnel** is the **overlay** network; it's a virtual connection built on top of this OSPF underlay, utilizing the underlay's routing capabilities to transport encapsulated traffic between R1 and R3.

Note: This lab is an exercise in configuring and verifying various implementations of GRE tunnels and does not reflect networking best practices.

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Ensure that the routers have been erased and have no startup configurations. If you are unsure contact your instructor.

Required Resources

- 3 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 1 PC (Choice of operating system with a terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet cables as shown in the topology

Instructions

Part 1: Build the Network and Configure Basic Device Settings

In Part 1, you will set up the network topology and configure basic settings.

Step 1: Cable the network as shown in the topology.

Attach the devices as shown in the topology diagram, and cable as necessary.

Step 2: Configure basic settings for each router.

a. Console into each router, enter global configuration mode, and apply the basic settings. A command list for each device is listed below for initial configurations.

Router R1

```
hostname R1
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
no ip domain lookup
ipv6 unicast-routing
banner motd # R1, Implement a GRE Tunnel #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
router ospf 4
 router-id 1.1.1.4
 exit
ipv6 router ospf 6
 router-id 1.1.1.6
 exit
interface g0/0/0
 ip address 10.1.2.1 255.255.255.0
 ipv6 address fe80::1:1 link-local
 ipv6 address 2001:db8:acad:12::1/64
 no shutdown
 ip ospf 4 area 0
 ipv6 ospf 6 area 0
 exit
interface loopback 0
 ip address 192.168.1.1 255.255.255.0
 ipv6 address fe80::1:2 link-local
 ipv6 address 2001:db8:acad:1::1/64
 no shutdown
 ip ospf 4 area 0
 ipv6 ospf 6 area 0
 exit
interface loopback 1
 ip address 172.16.1.1 255.255.255.0
 ipv6 address fe80::1:3 link-local
 ipv6 address 2001:db8:acad:1721::1/64
 no shutdown
 exit
```

Router R2

```
hostname R2
no ip domain lookup
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
ipv6 unicast-routing
banner motd # R2, Implement a GRE Tunnel #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
router ospf 4
 router-id 2.2.2.4
 exit
ipv6 router ospf 6
 router-id 2.2.2.6
 exit
interface g0/0/0
 ip address 10.1.2.2 255.255.255.0
 ipv6 address fe80::2:1 link-local
 ipv6 address 2001:db8:acad:12::2/64
 no shutdown
 ip ospf 4 area 0
 ipv6 ospf 6 area 0
 exit
interface g0/0/1
 ip address 10.2.3.2 255.255.255.0
 ipv6 address fe80::2:2 link-local
 ipv6 address 2001:db8:acad:23::2/64
 no shutdown
 ip ospf 4 area 0
 ipv6 ospf 6 area 0
 exit
```

Router R3

```
hostname R3
no ip domain lookup
ipv6 unicast-routing
banner motd # R3, Implement a GRE Tunnel #
line con 0
 exec-timeout 0 0
 logging synchronous
 exit
line vty 0 4
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
 privilege level 15
 password cisco123
 exec-timeout 0 0
 logging synchronous
 login
 exit
router ospf 4
 router-id 3.3.3.4
 exit
ipv6 router ospf 6
 router-id 3.3.3.6
 exit
interface g0/0/0
 ip address 10.2.3.3 255.255.255.0
 ipv6 address fe80::3:1 link-local
 ipv6 address 2001:db8:acad:23::3/64
 no shutdown
 ip ospf 4 area 0
 ipv6 ospf 6 area 0
 exit
interface loopback 0
 ip address 192.168.3.1 255.255.255.0
 ipv6 address fe80::3:2 link-local
 ipv6 address 2001:db8:acad:3::1/64
 no shutdown
 ip ospf 4 area 0
 ipv6 ospf 6 area 0
 exit
interface loopback 1
 ip address 172.16.3.1 255.255.255.0
 ipv6 address fe80::3:3 link-local
 ipv6 address 2001:db8:acad:1723::1/64
 no shutdown
 exit
```

b. Set the clock on each device to UTC time.

c. Save the running configuration to startup-config.

Part 2: Configure and Verify GRE Tunnels with Static Routing

In Part 2, you will configure and verify GRE tunnels between R1 and R3. You will use static routes for overlay reachability and dynamic routing for underlay reachability. You will create two tunnels: one for IPv4 and one for IPv6. GRE tunnels are highly flexible, with many implementation options beyond this lab.

Step 1: Verify reachability between R1 and R3.

a. From R1, ping R3 interface Loopback 0 using IPv4. All pings should be successful.

```
R1# ping 192.168.3.1
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

b. From R1, ping R3 interface Loopback 0 using IPv6. All pings should be successful.

```
R1# ping 2001:db8:acad:3::1
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3::1, timeout is 2 seconds:
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/2/7 ms
```

Step 2: Create an IPv4-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 0, specifying the IP address 100.100.100.1/30, a tunnel source of Loopback0, and a tunnel destination of 192.168.3.1. This sets up the GRE overlay endpoint on R1. The configured IPv4 address identifies the tunnel interface, the source uses R1's Loopback0, and the destination points to R3's underlay address.

```
R1(config)# interface tunnel 0
*Oct  2 14:06:29.376: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to down
R1(config-if)# ip address 100.100.100.1 255.255.255.252
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 192.168.3.1
*Oct  2 14:08:44.437: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
R1(config-if)# exit
```

b. On R1, create a static route to 172.16.3.0/24 via interface Tunnel 0. This directs overlay traffic for the 172.16.3.0/24 network through the GRE tunnel, ensuring reachability to R3's LAN via the overlay.

```
R1(config)# ip route 172.16.3.0 255.255.255.0 tunnel 0
```

c. On R3, create interface Tunnel 0, specifying the IP address 100.100.100.2/30, a tunnel source of Loopback0, and a tunnel destination of 192.168.1.1. This mirrors the GRE overlay configuration on R3. The IPv4 address assigns the peer tunnel interface, using R3's Loopback0 as the source and R1's underlay address as the destination.

```
R3(config)# interface tunnel 0
*Oct  2 14:19:52.135: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to down
R3(config-if)# ip address 100.100.100.2 255.255.255.252
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 192.168.1.1
*Oct  2 14:20:25.079: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
R3(config-if)# exit
```

d. On R3, create a static route to 172.16.1.0/24 via interface Tunnel 0. This sends overlay traffic for the 172.16.1.0/24 network through the GRE tunnel, providing reachability to R1's LAN via the overlay.

```
R3(config)# ip route 172.16.1.0 255.255.255.0 tunnel 0
```

e. On R1, issue the command **show interface tunnel 0** and examine the output.

```
R1# show interface tunnel 0
Tunnel0 is up, line protocol is up
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
  Hardware is Tunnel
  Internet address is 100.100.100.1/30
  MTU 9976 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 192.168.1.1 (Loopback0), destination 192.168.3.1
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 1 member (includes iterators), on
interface <OK>
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input never, output never, output hang never
  Last clearing of "show interface" counters 00:02:45
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     0 packets input, 0 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     0 packets output, 0 bytes, 0 underruns
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

f. From R1, **ping 172.16.3.1**. The pings should be successful.

Step 3: Create an IPv6-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 1, specifying the IPv6 address 2001:db8:ffff::1/64, a tunnel source of Loopback0, a tunnel destination of 2001:db8:acad:3::1, and the tunnel mode GRE IPv6.

```
R1(config)# interface tunnel 1
*Oct  2 14:20:34.159: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to down
R1(config-if)# ipv6 address 2001:db8:ffff::1/64
R1(config-if)# tunnel source loopback 0
*Oct  2 14:20:54.185: %LINEPROTO-5-UPDOWN: Line protocol on Interface Loopback9,
changed state to up
R1(config-if)# tunnel destination 2001:db8:acad:3::1
R1(config-if)# tunnel mode gre ipv6
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
R1(config-if)# exit
```

b. On R1, create a static route to 2001:db8:acad:1723::/64 via interface Tunnel 1.

```
R1(config)# ipv6 route 2001:db8:acad:1723::/64 tunnel 1
```

c. On R3, create interface Tunnel 1, specifying the IPv6 address 1002:db8:ffff::2/64, a tunnel source of Loopback0, and a tunnel destination of 2001:db8:acad:1::1.

```
R3(config)# interface tunnel 1
*Oct  2 14:28:35.629: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to down
R3(config-if)# ipv6 address 2001:db8:ffff::2/64
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 2001:db8:acad:1::1
R3(config-if)# tunnel mode gre ipv6
*Oct  2 14:29:36.529: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to up
R3(config-if)# exit
```

d. On R3, create a static route to 2001:db8:acad:1721::/64 via interface Tunnel 1.

```
R3(config)# ipv6 route 2001:db8:acad:1721::/64 tunnel 1
```

e. On R1, issue the command **show interface tunnel 1** and examine the output.

```
R1# show interface tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  MTU 1456 bytes, BW 100 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 255/255, rxload 255/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 2001:DB8:ACAD:1::1 (Loopback0), destination 2001:DB8:ACAD:3::1
   Tunnel Subblocks:
      src-track:
         Tunnel1 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 2 members (includes iterators),on
interface <OK>
  Tunnel protocol/transport GRE/IPv6
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Path MTU Discovery, ager 10 mins, min MTU 1280
  Tunnel transport MTU 1456 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:31, output 00:01:01, output hang never
  Last clearing of "show interface" counters 00:06:58
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 367000 bits/sec, 395 packets/sec
  5 minute output rate 367000 bits/sec, 395 packets/sec
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
246335 packets input, 28574884 bytes, 0 no buffer
Received 0 broadcasts (0 IP multicasts)
0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
246336 packets output, 28575000 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 unknown protocol drops
0 output buffer failures, 0 output buffers swapped out
```

f. From R1, **ping 2001:db8:acad:1723::1**. The pings should be successful.

Part 3: Configure and verify GRE Tunnels with Dynamic Routing

In Part 3, you will rebuild GRE tunnels between R1 and R3 using dynamic routing over the overlay and static routing in the underlay. The goal is to:

- Use static routes in the underlay so each router can reach the other's loopback addresses (tunnel endpoints).
- Run OSPF across the GRE tunnels to dynamically learn and advertise the LAN prefixes behind R1 and R3.
- Configure two tunnels: one for IPv4 and one for IPv6, and verify neighbor formation, routing table entries, and end-to-end reachability.

Step 1: Remove the Tunnel 0 and Tunnel 1 interfaces on R1 and R3.

Removing the existing tunnel interfaces ensures there are no residual configurations that could interfere with the new dynamic routing design. This clears prior tunnel states and any overlay-specific settings.

Issue the command **no interface tunnel 0** and **no interface tunnel 1** on R1 and R3.

Step 2: Replace the OSPF configuration on R1, R2, and R3 with static routing.

This step prepares the underlay so that each device can reach the other devices' loopbacks using static routes. The GRE tunnels will then form over these reachable endpoints, and OSPF will operate across the overlay.

a. On R1, R2, and R3, remove OSPF with the **no router ospf 4** and **no ipv6 router ospf 6** commands.

   This cleanly removes existing OSPFv2 and OSPFv3 processes to prevent unintended adjacencies or learned routes while you build the new overlay.

b. On R1 and R3, create IPv4 and IPv6 static default routes that point to R2.

c. On R2, create IPv4 and IPv6 static routes that point to R1 and R3 loopback 0 networks.

```
R2(config)# ip route 192.168.1.0 255.255.255.0 10.1.2.1
R2(config)# ip route 192.168.3.0 255.255.255.0 10.2.3.3
R2(config)# ipv6 route 2001:db8:acad:1::/64 2001:db8:acad:12::1
R2(config)# ipv6 route 2001:db8:acad:3::/64 2001:db8:acad:23::3
```

d. Verify that R1 can reach Loopback 0 on R3 with pings using a source address of the R1 Loopback 0 address.

```
R1# ping 192.168.3.1 source loopback 0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.3.1, timeout is 2 seconds:
Packet sent with a source address of 192.168.1.1
!!!!!
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/2 ms

R1# ping 2001:db8:acad:3::1 source loopback 0
Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 2001:DB8:ACAD:3::1, timeout is 2 seconds:
Packet sent with a source address of 2001:DB8:ACAD:1::1
!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/1/1 ms
```

Step 3: Create an IPv4-based GRE tunnel between R1 and R3.

This establishes the IPv4 GRE overlay for OSPFv2. The bandwidth and MTU settings help ensure accurate routing metrics and avoid fragmentation of encapsulated packets.

a. On R1, create interface Tunnel 0, specifying the IP address 100.100.100.1/30, bandwidth of 4000 kbps, a tunnel source of Loopback0, and a tunnel destination of 192.168.3.1.

- The Tunnel0 IP address defines the overlay adjacency point for OSPF.
- The bandwidth value informs routing and QoS features; it is not a hard rate limit.
- The **ip mtu 1400** accommodates GRE/IP overhead to reduce fragmentation risks on a 1500-byte underlay.
- The source and destination identify the underlay endpoints used for GRE encapsulation.

```
R1(config)# interface tunnel 0
R1(config-if)# ip address 100.100.100.1 255.255.255.252
R1(config-if)# bandwidth 4000
R1(config-if)# ip mtu 1400
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 192.168.3.1
*Oct  2 15:00:02.799: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed state to up
R1(config-if)# exit
```

b. On R1, configure OSPFv2 process-id 4 with router-id 1.1.1.4, and use network statements or interface configuration commands to include interface Tunnel 0 in Area 0 and interface Loopback 1 in Area 1. This will form the OSPF adjacency across the tunnel (Area 0) and advertises the Loopback1 prefix from a separate area (Area 1).

```
R1(config)# router ospf 4
R1(config-router)# router-id 1.1.1.4
R1(config-router)# network 100.100.100.0 0.0.0.3 area 0
R1(config-router)# network 172.16.1.0 0.0.0.255 area 1
R1(config-router)# exit
```

c. On R3, create interface Tunnel 0, specifying the IP address 100.100.100.2/30, bandwidth of 4000 kbps, a tunnel source of Loopback0, and a tunnel destination of 192.168.1.1. This mirrors the tunnel configuration on R1 so that the overlay is point-to-point between the two tunnel interface addresses.

```
R3(config)# interface tunnel 0
R3(config-if)# ip address 100.100.100.2 255.255.255.252
R3(config-if)# bandwidth 4000
R3(config-if)# ip mtu 1400
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 192.168.1.1
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
*Oct  2 15:06:18.091: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed
state to up
R3(config-if)# exit
```

d. On R3, configure OSPFv2 process-id 4 with router-id 3.3.3.4, and use network statements or interface configuration commands to include interface Tunnel 0 in Area 0 and interface Loopback 1 in Area 1. You should see an informational message that R3 has established full adjacency with R1.

```
R3(config)# router ospf 4
R3(config-router)# router-id 3.3.3.4
R3(config-router)# network 100.100.100.0 0.0.0.3 area 0
R3(config-router)# network 172.16.3.0 0.0.0.255 area 1
*Oct  2 15:06:18.108: %OSPF-5-ADJCHG: Process 4, Nbr 1.1.1.4 on Tunnel0 from LOADING
to FULL, Loading Done
R3(config-router)# exit
```

e. On R1, issue the command **show interface tunnel 0** and examine the output.

```
R1# show interface tunnel 0
Tunnel0 is up, line protocol is up
  Hardware is Tunnel
  Internet address is 100.100.100.1/30
  MTU 9976 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 192.168.1.1 (Loopback0), destination 192.168.3.1
   Tunnel Subblocks:
      src-track:
         Tunnel0 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 1 member (includes iterators), on
interface <OK>
  Tunnel protocol/transport GRE/IP
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255, Fast tunneling enabled
  Tunnel transport MTU 1476 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:01, output 00:00:04, output hang never
  Last clearing of "show interface" counters 00:06:11
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     23 packets input, 2064 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     58 packets output, 6784 bytes, 0 underruns
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

f. On R1, issue the command **show ip route ospf** and verify that 172.16.3.0/24 appears in the routing table as an OSPF route.

```
R1# show ip route ospf | begin Gateway
Gateway of last resort is 10.1.2.2 to network 0.0.0.0

      172.16.0.0/16 is variably subnetted, 3 subnets, 2 masks
O IA     172.16.3.1/32 [110/26] via 100.100.100.2, 00:02:53, Tunnel0
```

g. From R1, ping 172.16.3.1. The pings should be successful.

Step 4: Create an IPv6-based GRE tunnel between R1 and R3.

a. On R1, create interface Tunnel 1, specifying the IPv6 address 2001:db8:ffff::1/64, bandwidth of 4000kbps, a tunnel source of Loopback0, and a tunnel destination of 2001:db8:acad:3::1.

```
R1(config)# interface tunnel 1
R1(config-if)# ipv6 address 2001:db8:ffff::1/64
R1(config-if)# bandwidth 4000
R1(config-if)# tunnel source loopback 0
R1(config-if)# tunnel destination 2001:db8:acad:3::1
R1(config-if)# tunnel mode gre ipv6
*Oct  2 15:05:14.941: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to up
R1(config-if)# exit
```

b. On R1, configure OSPFv3 process-id 6 with router-id 1.1.1.6, and interface configuration commands to include interface Tunnel 0 in Area 0 and interface Loopback 1 in Area 1.

```
R1(config)# ipv6 router ospf 6
R1(config-rtr)# router-id 1.1.1.6
R1(config-rtr)# exit
R1(config)# interface tunnel 1
R1(config-if)# ipv6 ospf 6 area 0
R1(config-if)# exit
R1(config)# interface loopback 1
R1(config-if)# ipv6 ospf 6 area 1
R1(config-if)# exit
```

c. On R3, create interface Tunnel 1, specifying the IPv6 address 1002:db8:ffff::2/64, bandwidth of 4000kbps, a tunnel source of Loopback0, and a tunnel destination of 2001:db8:acad:1::1.

```
R3(config)# interface tunnel 1
R3(config-if)# ipv6 address 2001:db8:ffff::2/64
R3(config-if)# bandwidth 4000
R3(config-if)# tunnel source loopback 0
R3(config-if)# tunnel destination 2001:db8:acad:1::1
R3(config-if)# tunnel mode gre ipv6
*Oct  2 15:12:11.895: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel1, changed
state to up
R3(config-if)# exit
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

d. On R3, configure OSPFv3 process-id 6 with router-id 3.3.3.6, and use network statements or interface configuration commands to include interface Tunnel 0 in Area 0 and interface Loopback 1 in Area 1.

```
R3(config)# ipv6 router ospf 6
R3(config-rtr)# router-id 3.3.3.6
R3(config-rtr)# exit
R3(config)# interface tunnel 1
R3(config-if)# ipv6 ospf 6 area 0
R3(config-if)# exit
R3(config)# interface loopback 1
R3(config-if)# ipv6 ospf 6 area 1
*Oct  2 15:12:13.721: %OSPFv3-5-ADJCHG: Process 6, Nbr 1.1.1.6 on Tunnel1 from LOADING
to FULL, Loading Done
R3(config-if)# exit
```

e. On R1, issue the command **show interface tunnel 1** and examine the output.

```
R1# show interface tunnel 1
Tunnel1 is up, line protocol is up
  Hardware is Tunnel
  MTU 1456 bytes, BW 4000 Kbit/sec, DLY 50000 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation TUNNEL, loopback not set
  Keepalive not set
  Tunnel linestate evaluation up
  Tunnel source 2001:DB8:ACAD:1::1 (Loopback0), destination 2001:DB8:ACAD:3::1
   Tunnel Subblocks:
      src-track:
         Tunnel1 source tracking subblock associated with Loopback0
          Set of tunnels with source Loopback0, 2 members (includes iterators),on
interface <OK>
  Tunnel protocol/transport GRE/IPv6
    Key disabled, sequencing disabled
    Checksumming of packets disabled
  Tunnel TTL 255
  Path MTU Discovery, ager 10 mins, min MTU 1280
  Tunnel transport MTU 1456 bytes
  Tunnel transmit bandwidth 8000 (kbps)
  Tunnel receive bandwidth 8000 (kbps)
  Last input 00:00:09, output 00:00:04, output hang never
  Last clearing of "show interface" counters 00:04:20
  Input queue: 0/375/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/0 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     31 packets input, 4048 bytes, 0 no buffer
     Received 0 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
     46 packets output, 5864 bytes, 0 underruns
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
     0 output errors, 0 collisions, 0 interface resets
     0 unknown protocol drops
     0 output buffer failures, 0 output buffers swapped out
```

f. On R1, issue the command **show ipv6 route ospf** and verify that 2001:db8:acad:1723::/64 appears in the routing table as an OSPF route.

```
R1# show ipv6 route ospf | begin OI  2
OI  2001:DB8:ACAD:1723::1/128 [110/25]
     via FE80::12B3:D6FF:FE04:ED10, Tunnel1
```

g. From R1, ping 2001:db8:acad:1723::1. The pings should be successful.

Part 4: Examine the Recursive Routing Problem with GRE

Recursive routing occurs when a router chooses an overlay interface (the GRE tunnel) to reach the underlay next hop that the tunnel itself depends on. In other words, the router decides that the best path to the tunnel destination is via the tunnel. This creates a forwarding loop and causes the overlay to fail. In this part, you will intentionally trigger the issue to observe its symptoms and then correct the configuration to restore stability.

- Goal: Understand how advertising underlay prefixes into the overlay can cause recursion and how to avoid it.
- Symptom: OSPF adjacencies over the tunnel flap, and the platform logs a recursive routing error for the tunnel.
- Fix: Keep tunnel endpoint (underlay) reachability out of the overlay routing domain; use static routes or carefully filtered dynamic routing for the underlay.

a. To demonstrate how easily this could occur, add network 192.168.1.0 to the OSPF configuration of R1.

```
R1(config)# router ospf 4
R1(config-router)# network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# end
R1#
*Oct  2 15:21:30.148: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Oct  2 15:22:03.242: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.4 on Tunnel0 from FULL to
DOWN, Neighbor Down: Dead timer expired
*Oct  2 15:22:39.735: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.4 on Tunnel0 from LOADING
to FULL, Loading Done
*Oct  2 15:23:19.932: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.4 on Tunnel0 from FULL to
DOWN, Neighbor Down: Dead timer expired
*Oct  2 15:23:50.739: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.4 on Tunnel0 from LOADING
to FULL, Loading Done
*Oct  2 15:24:30.739: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.4 on Tunnel0 from FULL to
DOWN, Neighbor Down: Dead timer expired
R1#
```

After the dead timer expires, the OSPF adjacency over Tunnel0 begins to flap because the control-plane hellos and data-plane traffic intermittently lose a valid underlay path when the route to the tunnel destination points back into the tunnel. The “Dead timer expired” message indicates OSPF stopped receiving hellos, consistent with a transient forwarding loop.

b. R1 shows that the dead timer expires and then the adjacency tries to reset. Now look at what is being logged at R3.

```
*Oct  2 15:27:28.714: %ADJ-5-PARENT: Midchain parent maintenance for IP midchain out
of Tunnel0 - looped chain attempting to stack
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public

```
*Oct  2 15:27:37.107: %TUN-5-RECURDOWN: Tunnel0 temporarily disabled due to recursive
routing
*Oct  2 15:27:38.109: %LINEPROTO-5-UPDOWN: Line protocol on Interface Tunnel0, changed
state to down
*Oct  2 15:27:38.112: %OSPF-5-ADJCHG: Process 4, Nbr 1.1.1.4 on Tunnel0 from FULL to
DOWN, Neighbor Down: Interface down or detached
```

c. As you can see, R3 recognizes the issue and even tells you there is a recursive routing problem. Fix this by removing the network statement on R1 and the tunnel will come back up.

```
R1(config)# router ospf 4
R1(config-router)# no network 192.168.1.0 0.0.0.255 area 0
R1(config-router)# end
*Oct  2 15:27:06.673: %SYS-5-CONFIG_I: Configured from console by console
R1#
*Oct  2 15:27:23.766: %OSPF-5-ADJCHG: Process 4, Nbr 3.3.3.4 on Tunnel0 from LOADING
to FULL, Loading Done
```

© 2020 - 2026 Cisco and/or its affiliates. All rights reserved. Cisco Public